V-Dem Data Lab

Exploring V-Dem and V-Party in R

[Your Names]

2026-10-16

Getting started

This file is a toolbox for your individual exploration. Run the examples, then change the countries, years or indicators to follow your own interests. The aim is to understand a measure, discover a pattern and develop a research question. You do not need to work through every example.

For the second half of the data lab session, spend about 35 minutes exploring and keep the final 10 minutes for discussion. Choose either the V-Dem or the V-Party example.

In RStudio, run a code chunk using its green arrow. In Positron, use the >Run Cell button above each code cell. **Render** in RStudio and **Preview** in Positron produces a document from the whole file. Write your observations in the spaces below.

Preparation

Install packages once

Run this chunk manually in the beginning of the session. It is excluded from rendering (`eval: false`). Installation requires an internet connection and may take a few moments. See the official package repository for installation guidance.

```
install.packages(c("tidyverse", "remotes"))
remotes::install_github("vdeminstitute/vdemdata")
```

Load packages and datasets

```
if (!require(pacman)) install.packages("pacman")
p_load(tidyverse, vdemdata)

vdem_data <- vdemdata::vdem
vparty_data <- vdemdata::vparty
codebook <- vdemdata::codebook #vdem codebook only, vparty codebook available:
```

```
# Record the installed package version.
packageVersion("vdemdata")
```

```
[1] '16.0'
```

V-Dem here uses **country-year** observations. V-Party uses **country-party-date** observations: do not assume that each party is observed every year. Check the coverage of your chosen sample before comparing it. The datasets have different time coverage and release schedules.

1. Inspect the data and find a measure

Basic tools

```
# Number of rows and columns
dim(vdem_data)
```

```
[1] 28092  4618
```

```
dim(vparty_data)
```

```
[1] 11898   384
```

```
# A small preview, rather than printing the entire dataset
vdem_data |> select(country_name, year, v2x_polyarchy) |> head()
```

```
  country_name year v2x_polyarchy
1       Mexico 1789         0.027
2       Mexico 1790         0.027
3       Mexico 1791         0.027
4       Mexico 1792         0.027
5       Mexico 1793         0.027
6       Mexico 1794         0.027
```

```
vparty_data |> select(country_name, year, v2paenname, v2paid) |> head()
```

```
  country_name year                        v2paenname v2paid
1       Mexico 1991 Party of the Democratic Revolution    216
2       Mexico 1994 Party of the Democratic Revolution    216
3       Mexico 1997 Party of the Democratic Revolution    216
```

```
4       Mexico 2000 Party of the Democratic Revolution    216
5       Mexico 2003 Party of the Democratic Revolution    216
6       Mexico 2006 Party of the Democratic Revolution    216
```

```
# Available years
range(vdem_data$year, na.rm = TRUE)
```

```
[1] 1789 2025
```

```
range(vparty_data$year, na.rm = TRUE)
```

```
[1] 1900 2019
```

For an interactive spreadsheet-style view, use `view()`. These commands are optional and excluded from rendering.

```
view(codebook)
view(vdem_data)
view(vparty_data)
```

Search the V-Dem codebook

```
# Search indicator names by keyword; replace the search term.
democracy_variables <- vdemdata::find_var("democracy")
democracy_variables |> select(tag, name) |> head(10)
```

```
               tag                                    name
1    e_boix_regime                       Democracy (BMR)
2      e_chga_demo                               Democracy
3          e_democ           Institutionalized democracy
4            e_uds     Unified democracy score posterior
5   e_v2x_delibdem  Deliberative democracy index ordinal
6    e_v2x_egaldem   Egalitarian democracy index ordinal
7     e_v2x_libdem       Liberal democracy index ordinal
8  e_v2x_partipdem Participatory democracy index ordinal
9  e_v2x_polyarchy     Electoral democracy index ordinal
10      v2cademmob            Mobilization for democracy
```

```r
# Ask about a specific variable using its name in quotation marks.
vdemdata::var_info("v2x_polyarchy")
```

```
$name
[1] "Electoral democracy index"

$vartype
[1] "D"

$tag
[1] "v2x_polyarchy"

$projectmanager
[1] "Jan Teorell"

$question
[1] "To what extent is the ideal of electoral democracy in its fullest sense
achieved?"

$clarification
[1] "The electoral principle of democracy seeks to embody the core value of
making rulers responsive to citizens, achieved through electoral competition
for the electorate's approval under circumstances when suffrage is extensive;
political and civil society organizations can operate freely; elections are
clean and not marred by fraud or systematic irregularities; and elections affect
the composition of the chief executive of the country. In between elections,
there is freedom of expression and an independent media capable of presenting
alternative views on matters of political relevance. In the V-Dem conceptual
scheme, electoral democracy is understood as an essential element of any other
conception of representative democracy --- liberal, participatory, deliberative,
egalitarian, or some other."

$responses
[1] NA

$scale
[1] "Interval, from low to high (0-1)."

$notes
[1] NA

$crosscoder_aggregation
```

```
[1] NA

$cy_aggregation
[1] "Day-weighted mean"

$datarelease
[1] "1-16. Release 1-5 used a different, preliminary aggregation formula."

$years
[1] "1789-2025"

$convergence
[1] NA
```

`find_var()` returns a table of codebook entries, not just variable names. `var_info()` expects a variable name, not a dataset. These helpers are documented for the V-Dem codebook. For V-Party, consult its separate codebook on the dataset page.

You can search column names in either dataset. This finds names, not definitions:

```
names(vdem_data)[str_detect(names(vdem_data), "^v2x_")] |> head(15)
```

```
 [1] "v2x_polyarchy"          "v2x_polyarchy_codelow"  "v2x_polyarchy_codehigh"
 [4] "v2x_polyarchy_sd"       "v2x_libdem"             "v2x_libdem_codelow"
 [7] "v2x_libdem_codehigh"    "v2x_libdem_sd"          "v2x_partipdem"
[10] "v2x_partipdem_codelow"  "v2x_partipdem_codehigh" "v2x_partipdem_sd"
[13] "v2x_delibdem"           "v2x_delibdem_codelow"   "v2x_delibdem_codehigh"
```

```
names(vparty_data)[str_detect(names(vparty_data), "vote|seat|corrupt")]
```

```
[1] "v2paseatshare"       "v2panumbseat"        "v2patotalseat"
[4] "v2pavote"            "ep_corrupt_salience"
```

Your notes: Which variable interests you? What does it measure? What do high and low values mean? What aspect of your interest does it leave out?

Write here.

2A. V-Dem: Compare countries over time

Select countries, years and columns

```
# Change these settings to explore your own cases.
chosen_countries <- c("Germany", "Hungary", "Poland")
first_year <- 1990
last_year <- 2024

democracy_sample <- vdem_data |>
  filter(country_name %in% chosen_countries,
         year >= first_year, year <= last_year) |>
  select(country_name, year, v2x_polyarchy, v2x_libdem)

head(democracy_sample)
```

```
  country_name year v2x_polyarchy v2x_libdem
1       Poland 1990         0.740      0.660
2       Poland 1991         0.866      0.783
3       Poland 1992         0.888      0.804
4       Poland 1993         0.887      0.808
5       Poland 1994         0.890      0.811
6       Poland 1995         0.889      0.810
```

```
# Check how many observations and missing values each country has.
democracy_sample |>
  group_by(country_name) |>
  summarise(observations = n(),
            missing_electoral = sum(is.na(v2x_polyarchy)))
```

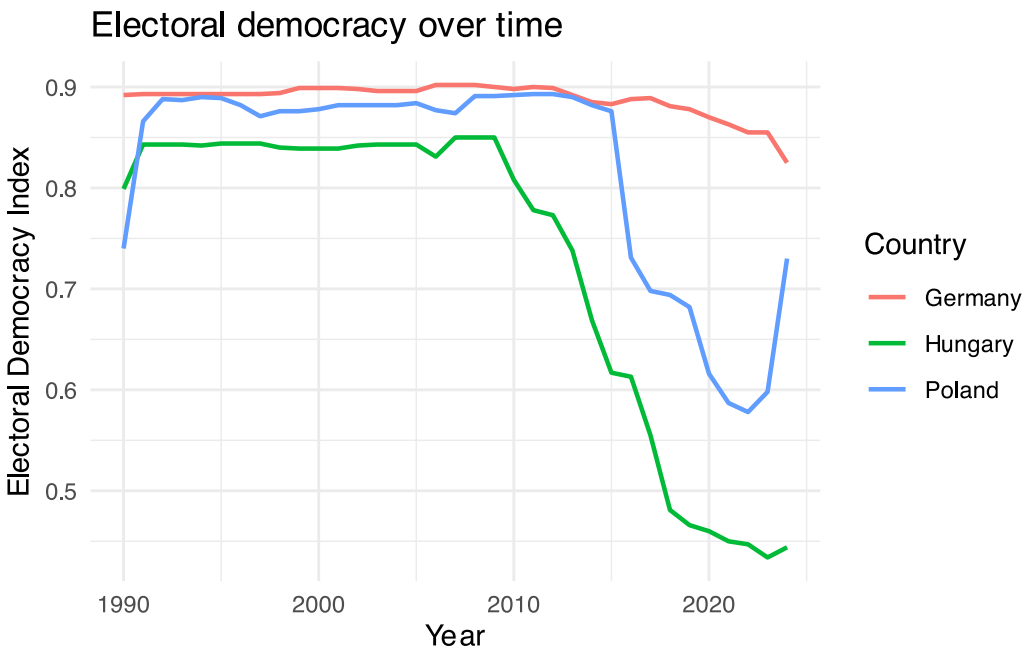
```
# A tibble: 3 × 3
  country_name observations missing_electoral
  <chr>               <int>             <int>
1 Germany                35                 0
2 Hungary                35                 0
3 Poland                 35                 0
```

`filter()` selects rows; `select()` selects columns. `%in%` matches any of the countries in your list. `|>` passes the result to the next step. `NA` means missing data, not zero.

Draw a line graph

```
democracy_plot <- ggplot(democracy_sample,
                         aes(x = year, y = v2x_polyarchy,
                             colour = country_name)) +
  geom_line(linewidth = 0.8) +
  labs(title = "Electoral democracy over time",
       x = "Year", y = "Electoral Democracy Index",
       colour = "Country") +
  theme_minimal()

democracy_plot
```




```
var_info("v2x_polyarchy")
```

```
$name
[1] "Electoral democracy index"

$vartype
[1] "D"

$tag
[1] "v2x_polyarchy"
```

```
$projectmanager
[1] "Jan Teorell"

$question
[1] "To what extent is the ideal of electoral democracy in its fullest sense
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$clarification
[1] "The electoral principle of democracy seeks to embody the core value of
making rulers responsive to citizens, achieved through electoral competition
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political and civil society organizations can operate freely; elections are
clean and not marred by fraud or systematic irregularities; and elections affect
the composition of the chief executive of the country. In between elections,
there is freedom of expression and an independent media capable of presenting
alternative views on matters of political relevance. In the V-Dem conceptual
scheme, electoral democracy is understood as an essential element of any other
conception of representative democracy --- liberal, participatory, deliberative,
egalitarian, or some other."

$responses
[1] NA

$scale
[1] "Interval, from low to high (0-1)."

$notes
[1] NA

$crosscoder_aggregation
[1] NA

$cy_aggregation
[1] "Day-weighted mean"

$datarelease
[1] "1-16. Release 1-5 used a different, preliminary aggregation formula."

$years
[1] "1789-2025"

$convergence
[1] NA
```

```r
var_info("v2x_libdem")
```

```
$name
[1] "Liberal democracy index"

$vartype
[1] "D"

$tag
[1] "v2x_libdem"

$projectmanager
[1] "Jan Teorell"

$question
[1] "To what extent is the ideal of liberal democracy achieved?"

$clarification
[1] "The liberal principle of democracy emphasizes the importance of protecting
individual and minority rights against the tyranny of the state and the tyranny
of the majority. The liberal model takes a \"negative\" view of political power
insofar as it judges the quality of democracy by the limits placed on government.
This is achieved by constitutionally protected civil liberties, strong rule of
law, an independent judiciary, and effective checks and balances that, together,
limit the exercise of executive power. To make this a measure of liberal
democracy, the index also takes the level of electoral democracy into account."

$responses
[1] NA

$scale
[1] "Interval, from low to high (0-1)."

$notes
[1] NA

$crosscoder_aggregation
[1] NA

$cy_aggregation
[1] "Day-weighted mean"
```

```
$datarelease
[1] "1-16. Release 1, 2, and 3 used a different, preliminary aggregation formula."

$years
[1] "1789-2025"

$convergence
[1] NA
```

Try it: Replace `v2x_polyarchy` in the plot with `v2x_libdem`, and adjust the title and axis label. Consult `var_info("v2x_libdem")` first. These indices capture different concepts; a difference between their values is not itself a measure of institutional change.

Your notes: Describe one pattern. Does it match your expectations? Separate the observation from your possible explanation.

Write here.

2B. V-Party: Compare parties at one election date

Party names are stored in `v2paenname` (English name), and party identifiers in `v2paid`. The example uses vote shares, `v2pavote`; consult the V-Party codebook for its definition, scale and treatment of electoral alliances.

```
chosen_country <- "Germany"

# Inspect available dates before choosing one.
vparty_data |>
  filter(country_name == chosen_country) |>
  distinct(year, historical_date) |>
  arrange(desc(year), historical_date) |>
  head(10)
```

```
   year historical_date
1  2017      2017-09-24
2  2013      2013-09-22
3  2009      2009-09-27
4  2005      2005-09-18
5  2002      2002-09-22
6  1998      1998-09-27
7  1994      1994-10-16
8  1990      1990-12-02
9  1987      1987-01-25
10 1983      1983-03-06
```

```
# Example: German parties observed in 2017.
party_sample <- vparty_data |>
  filter(country_name == chosen_country, year == 2017) |>
  select(country_name, year, historical_date,
         v2paid, v2paenname, v2pavote)

party_sample
```

```
  country_name year historical_date v2paid
1      Germany 2017      2017-09-24    211
2      Germany 2017      2017-09-24    383
3      Germany 2017      2017-09-24    573
4      Germany 2017      2017-09-24   1375
5      Germany 2017      2017-09-24   1545
6      Germany 2017      2017-09-24   1731
7      Germany 2017      2017-09-24   1816
8      Germany 2017      2017-09-24   1976
                                          v2paenname v2pavote
1 Christian Democratic Union / Christian Social Union       NA
2                  Social Democratic Party of Germany     20.5
3                              Free Democratic Party     10.7
4                          Christian Democratic Union     28.2
5                                           The Left      9.2
6                              Christian Social Union      6.5
7                                Alliance 90 / Greens      8.9
8                             Alternative for Germany     12.6
```

Check that your sample refers to one election date before making the chart. If your chosen year contains several dates, add a filter for `historical_date`. Missing parties or missing vote shares should not be interpreted as zero votes.

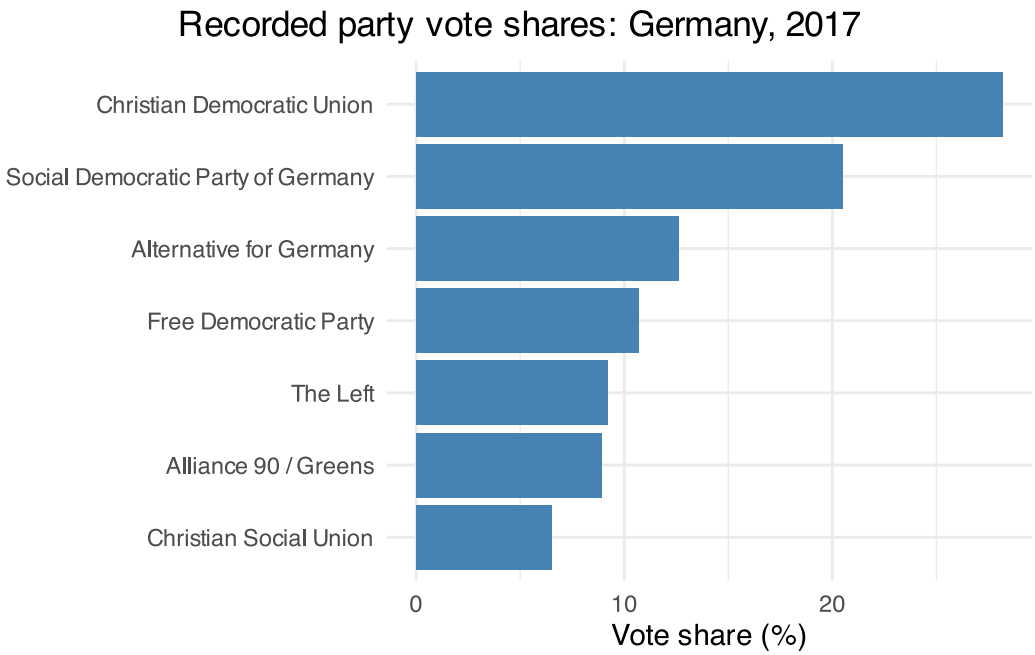
```
party_sample |> summarise(missing_vote_shares = sum(is.na(v2pavote)))
```

```
  missing_vote_shares
1                   1
```

```
party_plot <- ggplot(party_sample |> filter(!is.na(v2pavote)),
       aes(x = reorder(v2paenname, v2pavote), y = v2pavote)) +
        geom_col(fill = "steelblue") +
        coord_flip() +
        labs(title = "Recorded party vote shares: Germany, 2017",
```

```
            x = NULL, y = "Vote share (%)") +
        theme_minimal() +
        theme(
            plot.title.position = "plot",
            plot.title = element_text(hjust = 0.5)
        )

party_plot
```



Try it: Choose another country and an available election year. Update the chart title. Alternatively, find a party characteristic in the V-Party codebook, locate its column and explore it. Use party identifiers when tracking parties; names alone may not reliably identify them over time.

Your notes: What does your comparison show? What would you need to know about the election or party coverage to interpret it?

Write here.

3. Develop a research question

Based on your exploration, write down:

1. One research question.
2. The countries, parties and period you would investigate.
3. One additional variable or data source you would need.
4. Something your current graph cannot establish.

A descriptive pattern can inspire an explanation, but does not establish causality. Where uncertainty intervals are provided, consider them before interpreting small differences.

Write here.

Optional tools: Save your results

Run manually if you want to save your V-Dem sample and plot in the folder containing this file. These commands are excluded from rendering.

```
write_csv(democracy_sample, "data/my-vdem-sample.csv")

ggsave("output/my-democracy-plot.png", plot = democracy_plot,
       width = 8, height = 5, dpi = 300)

ggsave("output/my-party-plot.png", plot = party_plot,
       width = 8, height = 5, dpi = 300)
```

Sources and reproducibility

- vdemdata package documentation
- V-Dem data and codebook
- V-Party data and codebook

Record the package and dataset versions you use and cite the datasets in research. Values may be revised between releases: use one release throughout your comparison. HTML is the default output; select Typst when you want a PDF.